UNIT-II

DATA LINK LAYER

The data link layer in the OSI (Open System Interconnections) Model is in between the physical layer and the network layer. This layer converts the raw transmission facility provided by the physical layer to a reliable and error-free link.

Design issues of Data Link Layer

1. Service Provided to Network Layer
2. Framing
3. Error Control
4. Flow Control
5. Physical Addressing
6. Access Control

1. Service Provided to Network Layer

The types of services provided can be of three types

- Unacknowledged connectionless service
- Acknowledged connectionless service
- Acknowledged connection - oriented service

Unacknowledged connectionless service

- It consists of having the source machine send independent frames to destination machine without having the destination machine acknowledge them.

  Eg: Ethernet
- The service rate is used when the error rate is low.

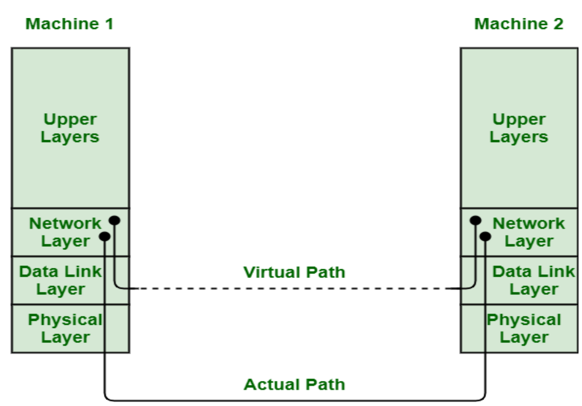
Acknowledged connectionless service

- There is no logical connections b/w the sender and receiver.
- Each and every frame sent is individually acknowledged
- This service simply provides acknowledged connectionless service i.e. packet delivery is simply acknowledged, with help of stop and wait for protocol.

Acknowledged connection - oriented service

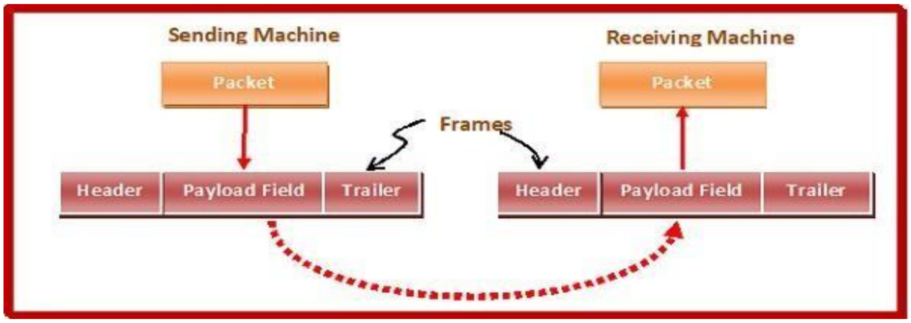
- In this type of service, connection is established first among sender and receiver or source and destination before data is transferred.
- It is guarantees that frame is received exactly once.

This process is shown in diagram



2.Framing

- The Data Link Layer should detect and correct the errors
- In this purpose, DLL will break up the bit stream into discrete frames, compute a small token called a checksum each frame and include the checksum in the frame when it is transmitted.
- When the frame arrives at the destination, the checksum is recomputed.
- After dividing the data into frames, we should be able to identify the starting & and ending of each frame.



Frames Parts:

- Frame Header – It contains the source and the destination addresses of the frame.
- Payload field – It contains the message to be delivered.
- Trailer – It contains the error detection and error correction bits.
- Flag – It marks the beginning and end of the frame.

There are two types of framing

1. Fixed-sized Framing:

Here the size of the frame is fixed and so the frame length acts as delimiter of the frame. Consequently, it does not require additional boundary bits to identify the start and end of the frame.

Example – ATM cells.

2. Variable Length: Here, the size of each frame to be transmitted may be different. So additional mechanisms are kept to mark the end of one frame and the beginning of the next frame.

It is used in local area networks.

It could be implemented in two different forms:

- **Length field**: In a frame, we can define the length field to show the length of a frame. It is applied in the Ethernet (802.3). The issue with it is that the length field may get corrupted sometimes.
- **ED (End Delimeter):** In a frame, we can address an end delimeter to show the completion of a frame. It is applied in the Token Ring. The issue with it is that the end delimeter can appear in the data.

There are four methods in the framing

- Character Count
- Flag Byte or Character Stuffing or Byte Stuffing
- Bit Stuffing
- Violation of Physical layer

Character Count or Byte Count:

- This is a method uses a field in the header to specify the number of bytesin the frame.
- When the DLL at the destination sees the byte count, it knows how manybytes follow & hence where the end of the frame.
- This problem occurs if the byte count is changed by any transmissionerror.
- Small example frames of sizes **5, 5, 8and 8 bytes**, respectively are shown.
- This problem occurs if the byte count is changed by any transmission error.
- If the byte count of 5 becomes 7due to error in this method is used rarely.

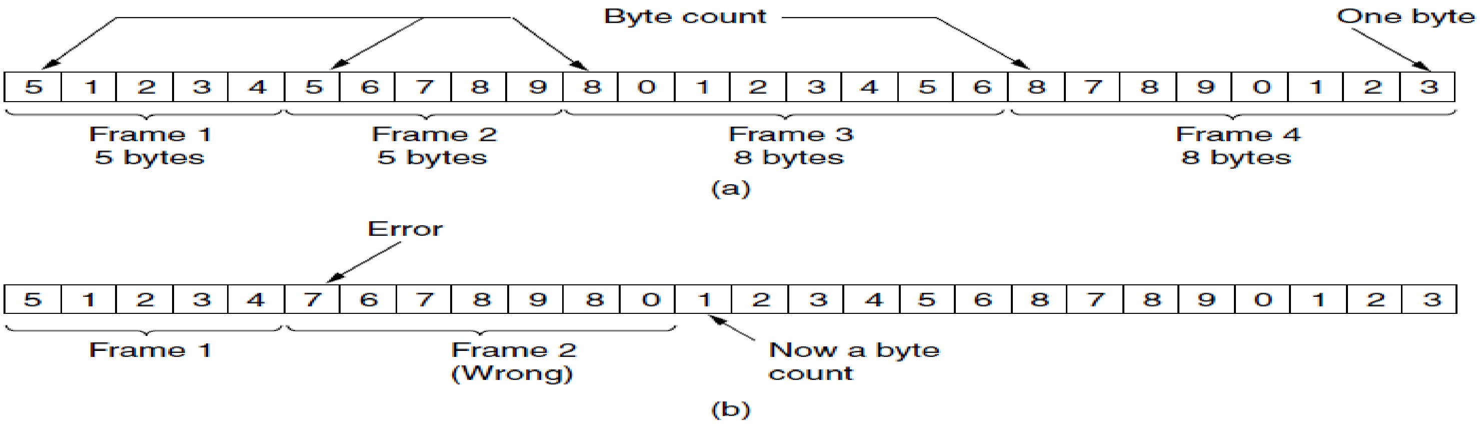


Figure 3-3. A byte stream. (a) Without errors. (b) With one error.

Flag Byte or Character Stuffing or Byte Stuffing

- In this method special byte called flag byte is used as both the starting & ending delimiter of each frame.
- Two consequent flag bytes indicate end of the frame and start the next frame.
- If the receiver looses the synchronization, it can search for two flag bytes to find the end of the frame.
- One way to solve this problem is to insert a special byte called (ESC) just before each flag byte in the data.
- The DLL on the receiving end removes the escape bytes before giving the data to the network layer.
- This technique is called as Byte Stuffing.

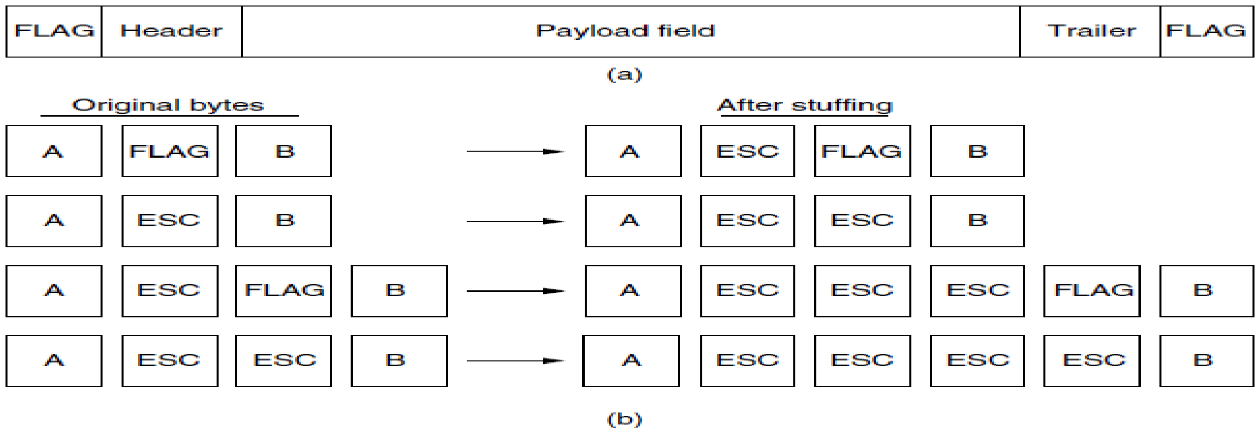


Figure 3-4. (a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.

Bit Stuffing

- Most protocols used a special 8bit pattern flag 01111110 as the delimiter the beginning and ending of the frame. The bit stuffing is done at the sender end and bit removal at the receiver end.
- If 0 after five consequent 1s we still stuff a 0.The receiver remove the 0.

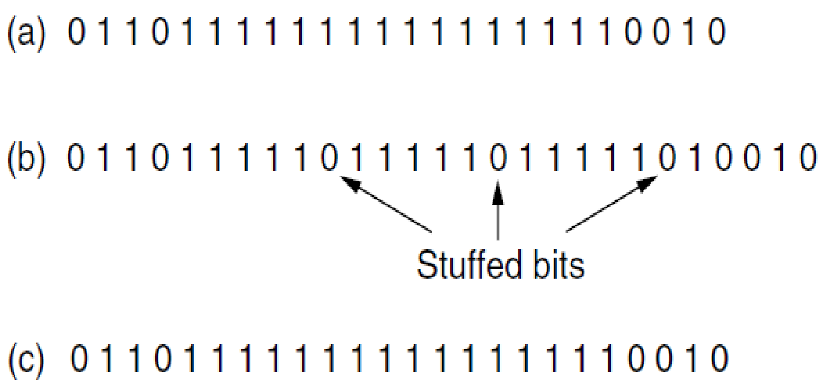


Figure 3-5. Bit stuffing. (a) The original data. (b) The data as they appear on the line. (c) The data as they are stored in the receiver's memory after destuffing.

Violation of Physical layer

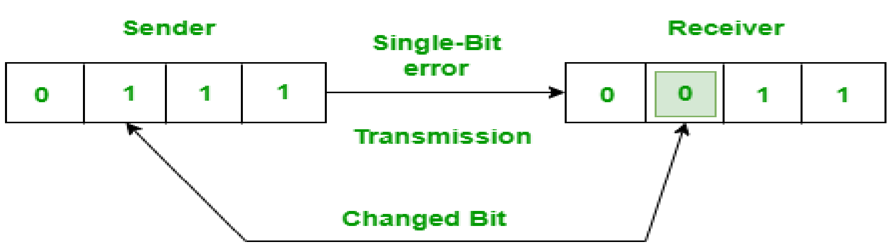
- In order to operate a division between frames in Data Link Layer this approach exploits the redundancy in Physical Layer Encoding that represents data as 00 error, 01 low, 10 high, 11 error

3.Error Control:

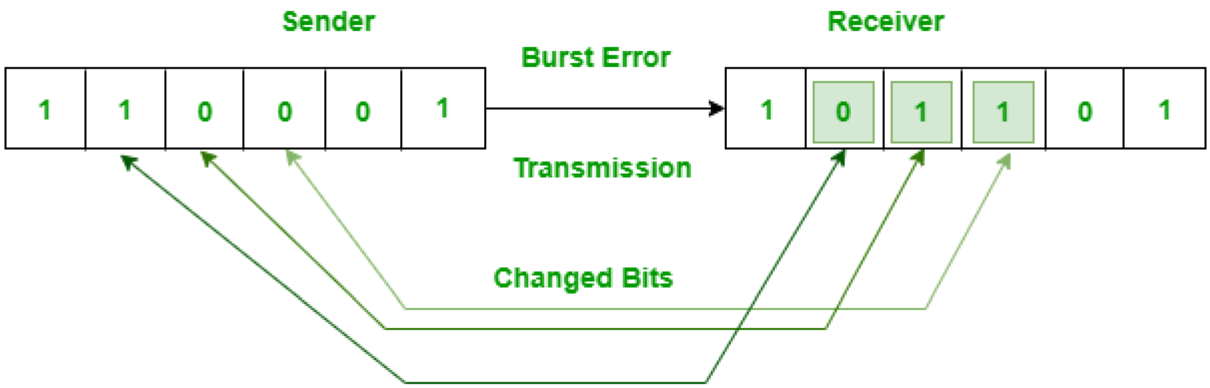
- Error control in data link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission.
- To ensure reliable delivery, the sender should be provided with some feedback about what is happening at receiver.
- In this purpose ,receiver sends special control frames having positive or negative acknowledgement
- If sender receives positive acknowledgment, it means that the frame has transmitted safely.
- If sender receives negative acknowledgment, it means that the frame is lost and the sender must retransmit the frame.
- To overcome this timer are used in DLL
- When sender transmits 0 frames, it also starts a timer.
- The timer is set to the time interval required for the data to reach the destination and ACK to reach the source.
- If the timers expire, it means that either the frame is lost or ACK is lost.

Types of Errors:

1. **Single-Bit Error:**
   if only one bit from this whole data packet is being changed/corrupted/altered then Single-bit transmission error occurs.

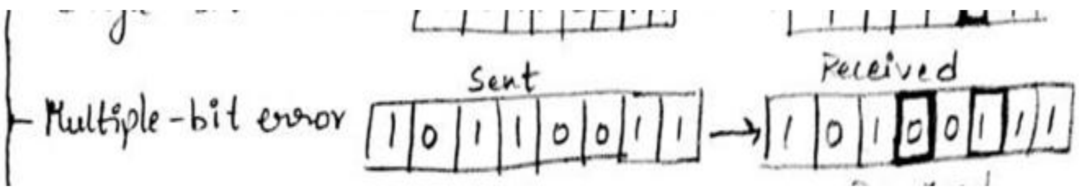


2. **Burst Error:**
   Two or more Consecutives multiple data bits of a data packets are being changed/corrupted/ altered during transmission



3. Multi bit Error:

Two or more Non-Consecutives multiple data bits of a data packets are being changed/corrupted/ altered during transmission



4. Flow control

Flow control is a set of procedures that restrict the amount of data a sender should send before it waits for some acknowledgment from the receiver.

- Flow Control is an essential function of the data link layer.
- It determines the amount of data that a sender can send.
- It makes the sender wait until an acknowledgment is received from the receiver's end.
- Methods of Flow Control are Stop-and-wait , and Sliding window.

Error detection and Corrections:

Error Control is a combination of both error detection and error correction. It ensures that the data received at the receiver end is the same as the one sent by the sender.

Error detection is the process by which the receiver informs the sender about any erroneous frame (damaged or lost) sent during transmission.

Error correction means it identifies the position of a error and it corrects the errors.

It refers to the retransmission of those frames by the sender.

Error detection codes:

1. Parity 2.Checksum 3.CRC (Cyclic Redundancy check)

1. **Parity**: It can detect single bit error.

There are two types: 1.Even parity 2. Odd parity

Data word Original data	even parity	Codeword Transmitted data	odd parity	Codeword transmitted data
1011010	0	10110100	1	10110101
100101	1	1001011	0	1001010

If transmitted data is | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | → parity bit

~~Bo~~ At the receiver,

If the received data is | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | → parity bit

Receiver calculates the parity bit

⇒ parity bit = 1 [Consider even parity]

the transmitted parity bit & receiver calculated parity bit are not equal. So, error is occured.

2. Checksum or 1's Representation Method

Step-01:

At sender side,

- If m bit checksum is used, the data unit to be transmitted is divided into segments of m bits.
- All the m bit segments are added.
- The result of the sum is then complemented using 1's complement arithmetic.
- The value so obtained is called as **checksum**.

Step-02:

- The data along with the checksum value is transmitted to the receiver.

Step-03:

At receiver side,

- If m bit checksum is being used, the received data unit is divided into segments of m bits.
- All the m bit segments are added along with the checksum value.
- The value so obtained is complemented and the result is checked.

Then, following two cases are possible-

Case-01: Result = 0

- If the result is zero,
- Receiver assumes that no error occurred in the data during the transmission.
- Receiver accepts the data.

Case-02: Result ≠ 0

- If the result is non-zero,
- Receiver assumes that error occurred in the data during the transmission.
- Receiver discards the data and asks the sender for retransmission.

Checksum Example-

Consider the data unit to be transmitted is-

10011001111000100010010010000100

Consider 8 bit checksum is used.

Step-01:

At sender side,

The given data unit is divided into segments of 8 bits as-

10011001	11100010	00100100	10000100

Now, all the segments are added and the result is obtained as-

- 10011001 + 11100010 + 00100100 + 10000100 = 1000100011
- Since the result consists of 10 bits, so extra 2 bits are wrapped around.
- 00100011 + 10 = 00100101 (8 bits)
- Now, 1's complement is taken which is 11011010.
- Thus, checksum value = 11011010

Step-02:

The data along with the checksum value is transmitted to the receiver.

Step-03:

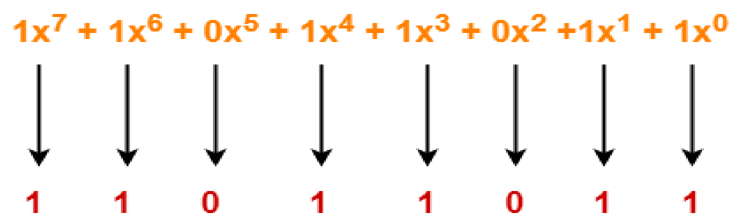
At receiver side,

- The received data unit is divided into segments of 8 bits.
- All the segments along with the checksum value are added.
- Sum of all segments + Checksum value = 00100101 + 11011010 = 11111111
- Complemented value = 00000000

Since the result is 0, receiver assumes no error occurred in the data and therefore accepts it.

3. CRC (Cyclic Redundancy Check)

- Cyclic Redundancy Check (CRC) is an error detection method.
- It is based on binary division.
- CRC generator is an algebraic polynomial represented as a bit pattern.
- Bit pattern is obtained from the CRC generator using the following rule is The power of each term gives the position of the bit and the coefficient gives the value of the bit.
- The polynomial code bit strings are representation of polynomials with coefficients of 0 and 1 only.
- When the polynomial code is employed, the sender and receiver must agree upon a generator polynomial G(x).
- The result is 0 check summed frame to be transmitted T(X).



Here the Binary pattern is 11011011

CRC Generator

If CRC generator or divisor is n bits the CRC bits are (n-1)

Formula for CRC bits =data+ (n-1)/CRC Generator

Ex:- Frame $M(x)$ = 1101011111

Generator $G(x)$ = 10011

code to be appended at the end of $M(x)$

Sender side = [(No of bits in $G(x)$) − 1] = 5 − 1 = 4 (0000)

```
10011 ) 11010111110000 (110000111
        10011
        -----
         10011
         10011
         -----
          00001
          00000
          -----
           00011
           00000
           -----
            00111
            00000
            -----
             01111
             00000
             -----
              11110
              10011
              -----
               11010
               10011
               -----
                10010
                10011
                -----
                 00010
                 00000
                 -----
                  0010
```

Transmitted frame $T(x)$ =

11010111110010

Receiver side :

```
10011 ) 11010111110010 ( 1100001110
        10011
         10011
         10011
          00001
          00000
           00011
           00000
            00111
            00000
             01111
             00000
              11110
              10011
               11010
               10011
                10011
                10011
                 00000
                 00000
                   (0)
```

At the receiver, the above calculation is done.
If the remainder is 'o'. It means there is no error.

Error Detection and Correction Method-Hamming Code

Hamming code is a block code that is capable of detecting up to two simultaneous bit errors and correcting single-bit errors. It was developed by **R.W. Hamming** for error correction.

In this coding method, the source encodes the message by inserting redundant bits within the message. These redundant bits are extra bits that are generated and inserted at specific positions in the message itself to enable error detection and correction. When the destination receives this message, it performs recalculations to detect errors and find the bit position that has error.

Encoding a message by Hamming Code (Sender)

The procedure used by the sender to encode the message encompasses the following steps −

Step 1 − Calculation of the number of redundant bits.
Step 2 − Positioning the redundant bits.
Step 3 − Calculating the values of each redundant bit.

Once the redundant bits are embedded within the message, this is sent to the user.

Step 1 − Calculation of the number of redundant bits.

If the message contains mmnumber of data bits, rrnumber of redundant bits are added to it so that mr is able to indicate at least $(m + r+ 1)$ different states. Here, $(m + r)$ indicates location of an error in each of $(+ r)$ bit positions and one additional state indicates no error. Since, rr bits can indicate $2^r r$ states, $2^r r$ must be at least equal to $(m + r + 1)$. Thus the following equation should hold 2r ≥ m+r+1

Step 2 − Positioning the redundant bits.

The r redundant bits placed at bit positions of powers of 2, i.e. 1, 2, 4, 8, 16 etc. They are referred in the rest of this text as r_1 (at position 1), r_2 (at position 2), r_3 (at position 4), r_4 (at position 8) and so on.

Step 3 − Calculating the values of each redundant bit.

The redundant bits are parity bits. A parity bit is an extra bit that makes the number of 1s either even or odd. The two types of parity are −

Even Parity – Here the total number of bits in the message is made even.
Odd Parity – Here the total number of bits in the message is made odd

Decoding a message in Hamming Code (Receiver)

Once the receiver gets an incoming message, it performs recalculations to detect errors and correct them. The steps for recalculation are –

Step 1 – Calculation of the number of redundant bits.
Step 2 – Positioning the redundant bits.
Step 3 – Parity checking.
Step 4 – Error detection and correction

Ex: Let us assume the even parity hamming code from the above example (111001101) is transmitted and the received code is (110001101). Now from the received code, let us detect and correct the error.

To detect the error, let us construct the bit location table.

Bit Location	9	8	7	6	5	4	3	2	1
Bit designation	D5	**P4**	D4	D3	D2	**P3**	D1	**P2**	**P1**
Binary representation	1001	1000	0111	0110	0101	0100	0011	0010	0001
Received code	1	1	0	0	0	1	1	0	1

Checking the parity bits

For P1 : Check the locations 1, 3, 5, 7, 9. There is three 1s in this group, which is wrong for even parity. Hence the bit value for P1 is 1.

For P2: Check the locations 2, 3, 6, 7. There is one 1 in this group, which is wrong for even parity. Hence the bit value for P2 is 1.

For P3 : Check the locations 3, 5, 6, 7. There is one 1 in this group, which is wrong for even parity. Hence the bit value for P3 is 1.

For P4 : Check the locations 8, 9. There are two 1s in this group, which is correct for even parity. Hence the bit value for P4 is 0.

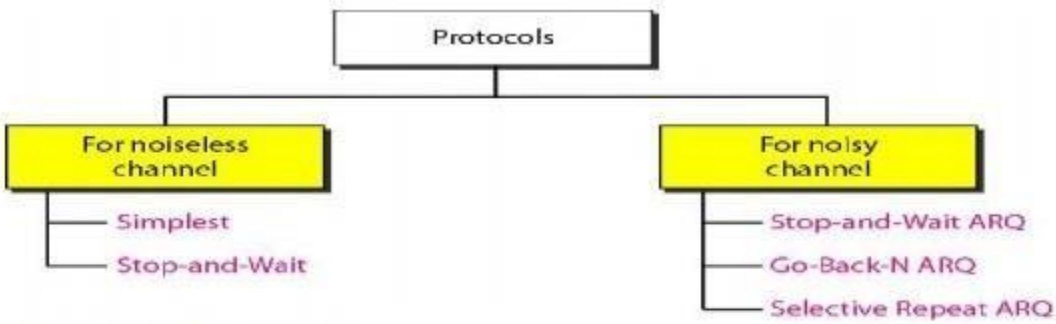
The resultant binary word is 0111. It corresponds to the bit location 7 in the above table. The error is detected in the data bit D4. The error is 0 and it should be changed to 1. **Thus the corrected code is 111001101**.

Elementary Data Link Layer Protocols

Based on error control and flow control mechanism the elementary data link layer protocols are divided in two transmission channels.

1. Noise Less

2. Noisy



Simplex Protocol or Unrestricted Protocol or Utopian Protocol:

a) Unrestricted Simplex protocol:

- It is also called as Utopian Simplex protocol.
- This protocol can be used if the following conditions exist :-

① Data are transmitted in one direction only.

② Both transmitting (sender) and receiving (receiver) systems are always ready

③ Processing time can be ignored.

④ Infinite buffer space is available

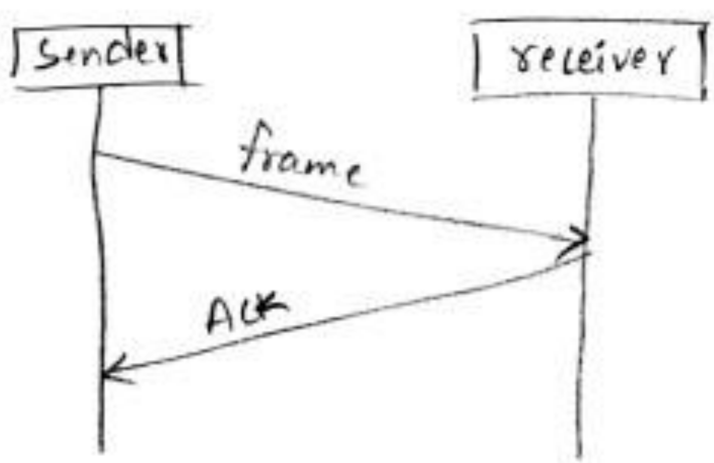
⑤ Communication channel never damages or looses frames.

⑥ No sequence numbers or Acknowledgements are used here.

- This protocol is unrealistic because it doesn't handle either flow control or error control.
- Its processing is close to that of an unacknowledged connectionless service.

Simplex Stop and Wait Protocol error free channel:

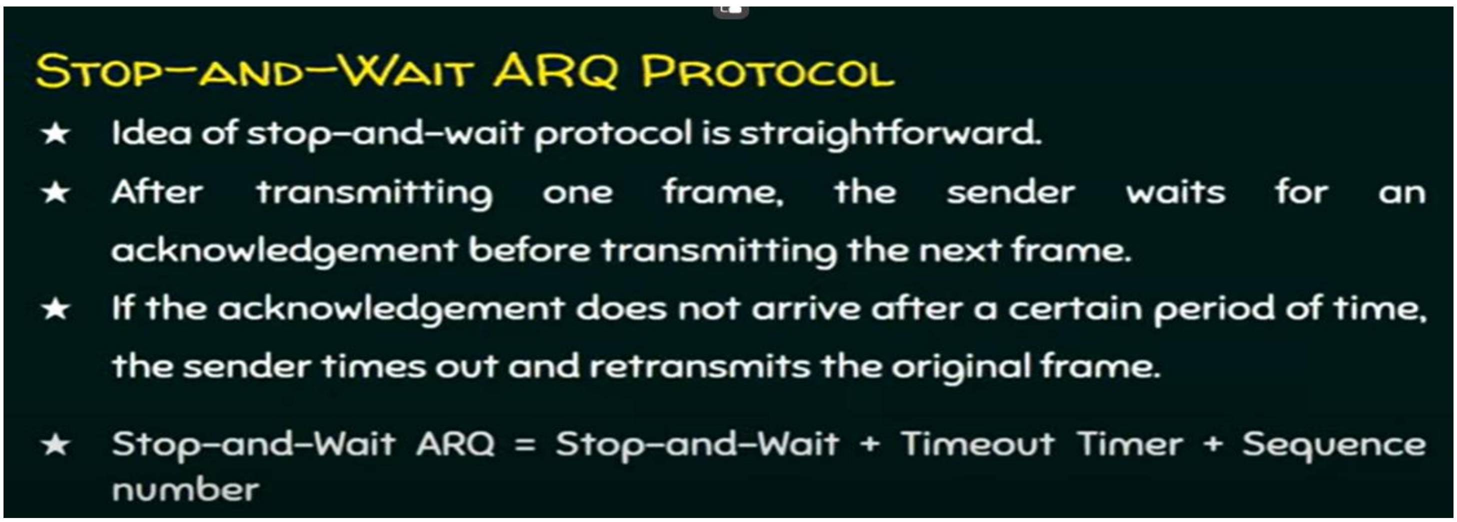
- The communication channel is assumed to be error free.
- The data traffic is half-duplex.
- In this protocol, receiver provides a feedback to the sender.
- It means that when the sender sends the data, the receiver receives it & sends a little dummy frame back to the sender giving permission to the sender to transmit the next frame.
- After having sent a frame, the sender is required by the protocol to wait until the dummy (ACK) frame arrives.
- Protocols in which the sender sends one frame & then waits for an ACK before proceeding to the next frame are called Stop-and-wait protocols.



Simplex stop and wait protocol for noisy channel:

A Noisy Channel Protocol is a type of communication protocol that is used in communication systems where the transmission channel may introduce errors into the transmitted data.
This type of protocol is designed to deal with errors in the communication channel and ensure that the data being transmitted is received accurately at the receiver end.
Some examples of Noisy Channel Protocols include the Stop-and-Wait Protocol, the Sliding Window Protocol, and the Automatic Repeat Request (ARQ) Protocol.

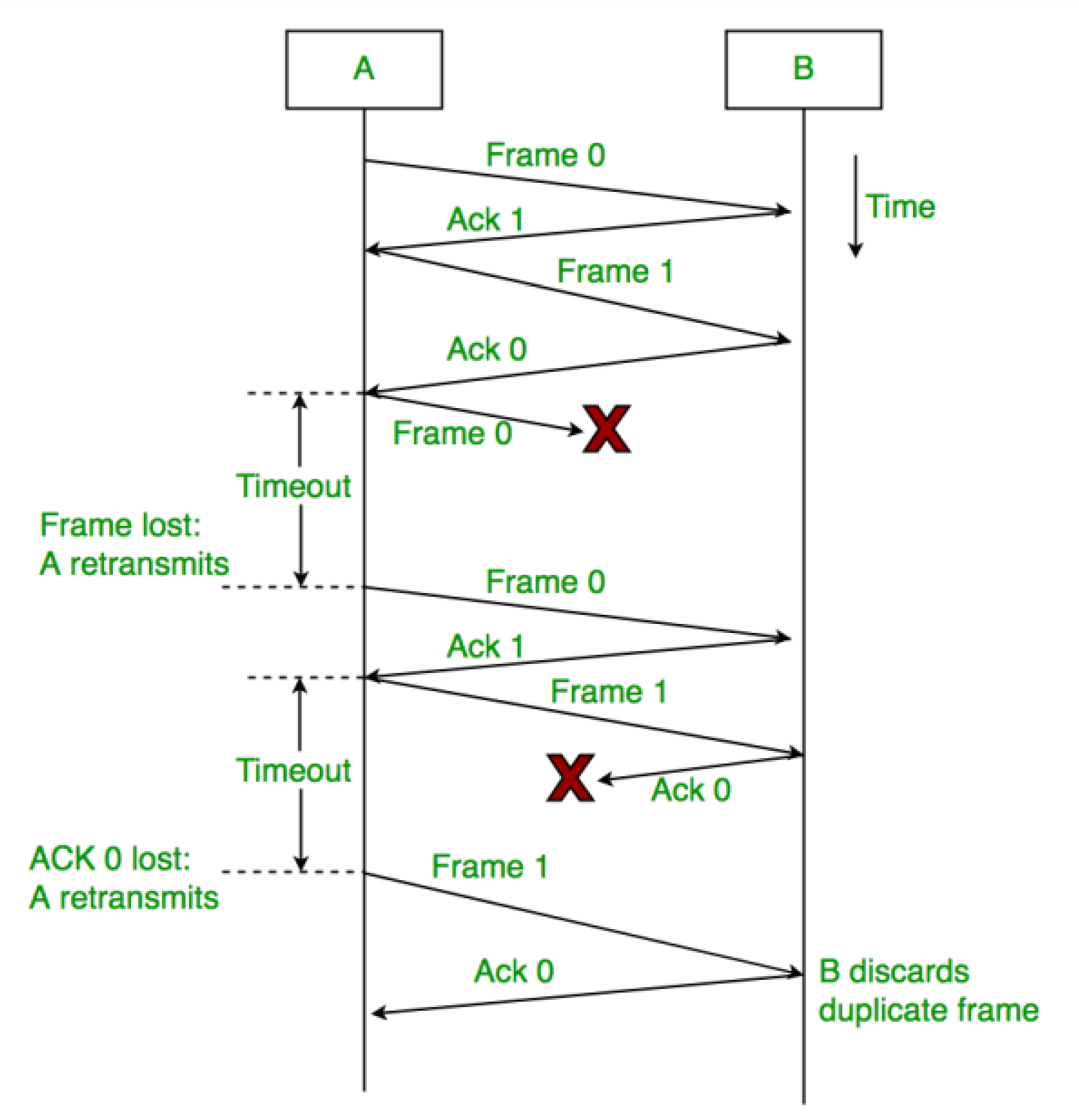
STOP-AND-WAIT ARQ Protocol:

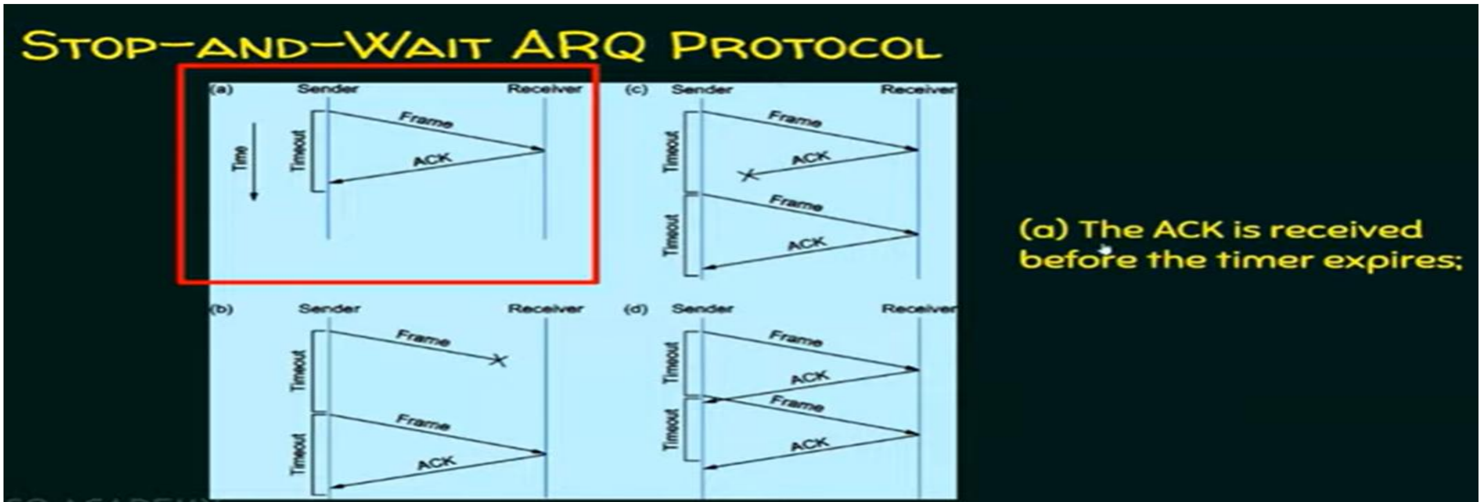


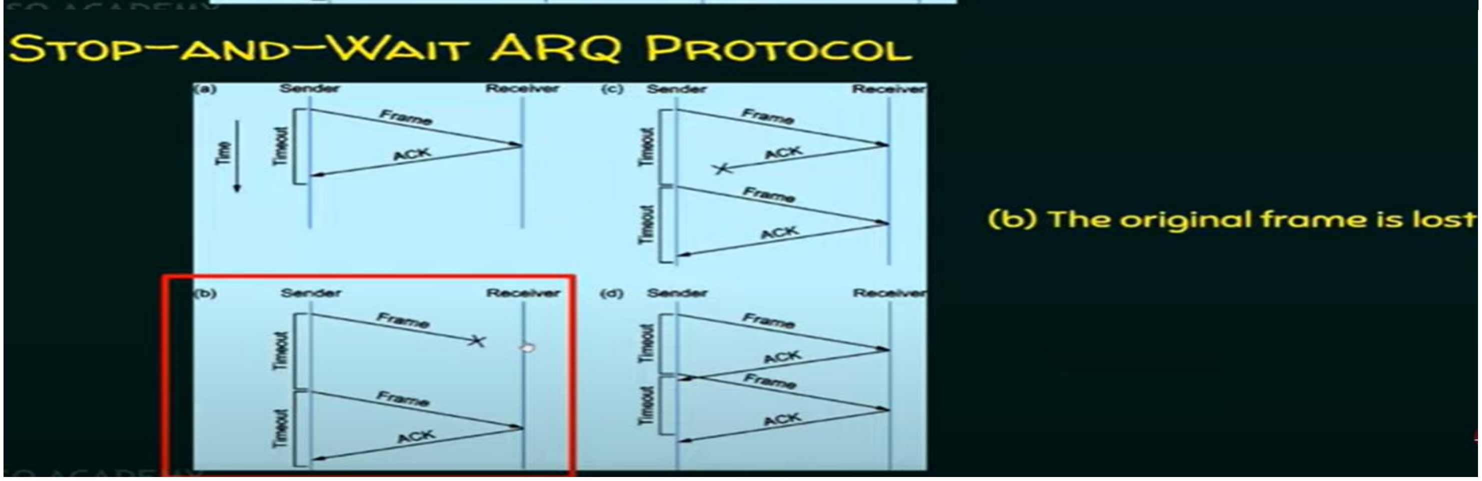
A data flow diagram in the Stop-and-Wait protocol in a noisy channel can be used to describe the flow of data between the sender and the receiver. This diagram generally includes the following components:

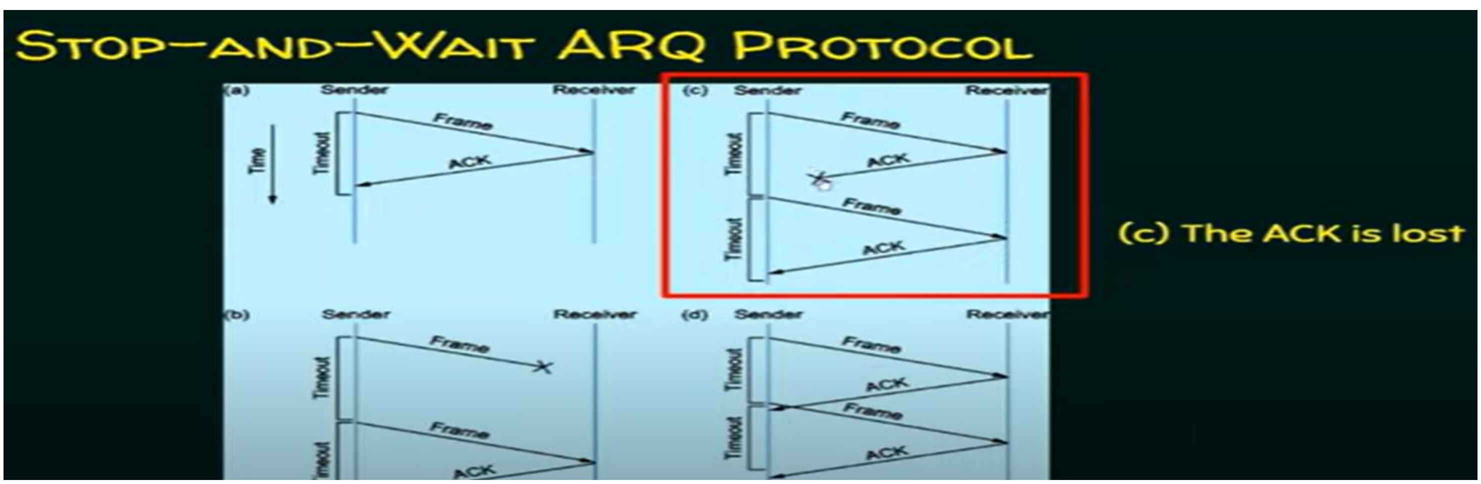
1. Sender: The sender sends data frames one at a time, and waits for a response (ACK or NACK) from the receiver before sending the next data frame.
2. Receiver: The receiver receives the data frames and processes them. If the frame is received correctly, the receiver sends an ACK signal to the sender. If the frame is not received correctly, the receiver sends a NACK signal to the sender.
3. Noisy Channel: The noisy channel is the medium through which the data frames are transmitted from the sender to the receiver. The channel can add noise to the data frames, resulting in errors and corruption of the data.
4. Error Detection: The receiver uses error detection techniques such as checksums to detect errors in the received data frames.

5. Error Correction: If an error is detected, the receiver sends a NACK signal to the sender, requesting a retransmission of the frame.



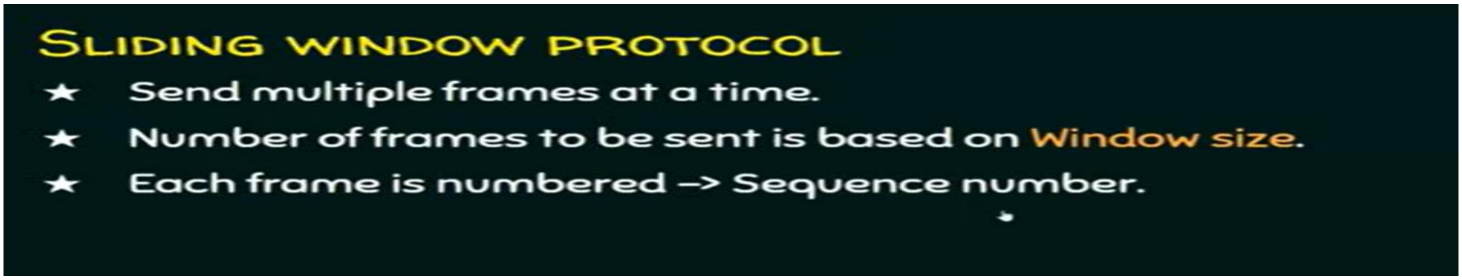

Stop-and-Wait ARQ Protocol
(a) Sender Receiver
Time
Timeout
Frame
ACK
(b) Sender Receiver
(c) Sender Receiver
(d) Sender Receiver
(a) The ACK is received before the timer expires;


Stop-and-Wait ARQ Protocol
(a) Sender Receiver
Time
Timeout
Frame
ACK
(b) Sender Receiver
(c) Sender Receiver
(d) Sender Receiver
(b) The original frame is lost


Stop-and-Wait ARQ Protocol
(a) Sender Receiver
Time
Timeout
Frame
ACK
(b) Sender Receiver
(c) Sender Receiver
(d) Sender Receiver
(c) The ACK is lost

Sliding window protocol:

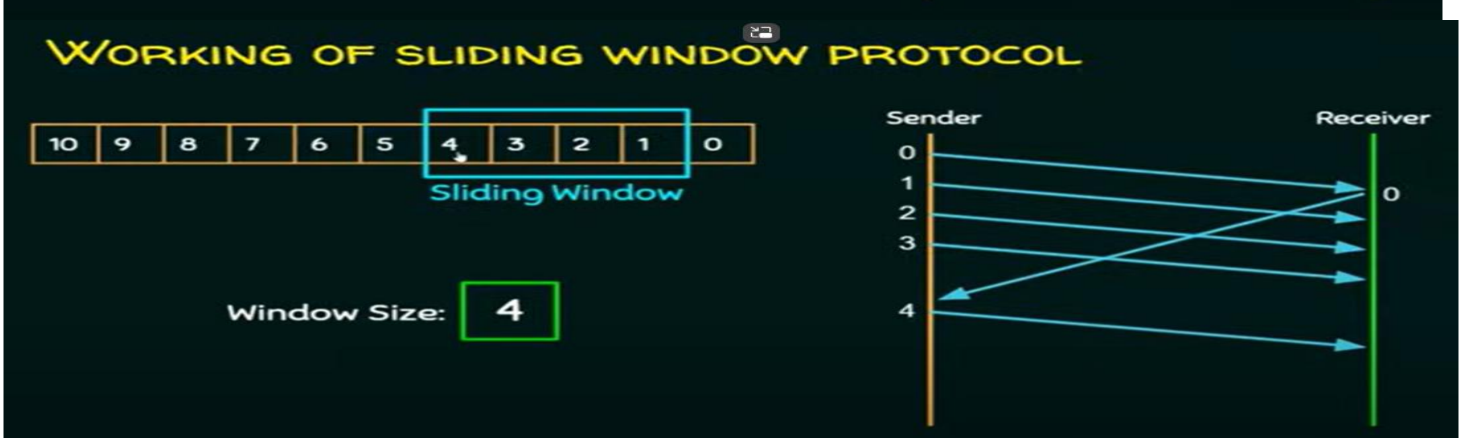
In this protocol, multiple frames can be sent by a sender at a time before receiving an acknowledgment from the receiver.

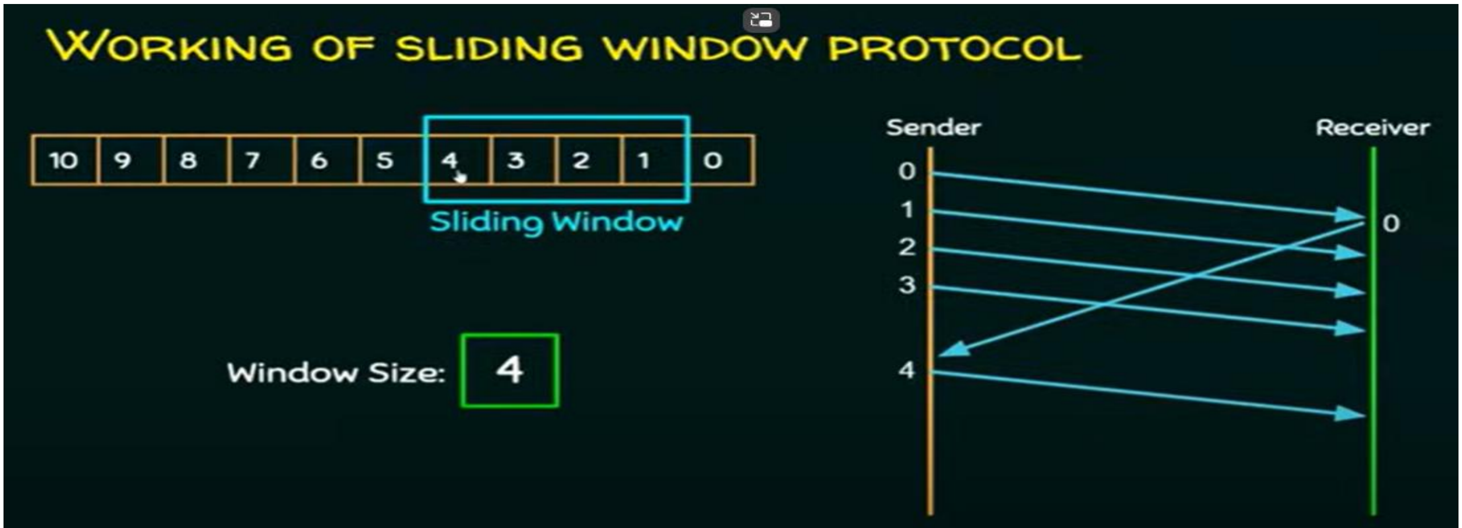


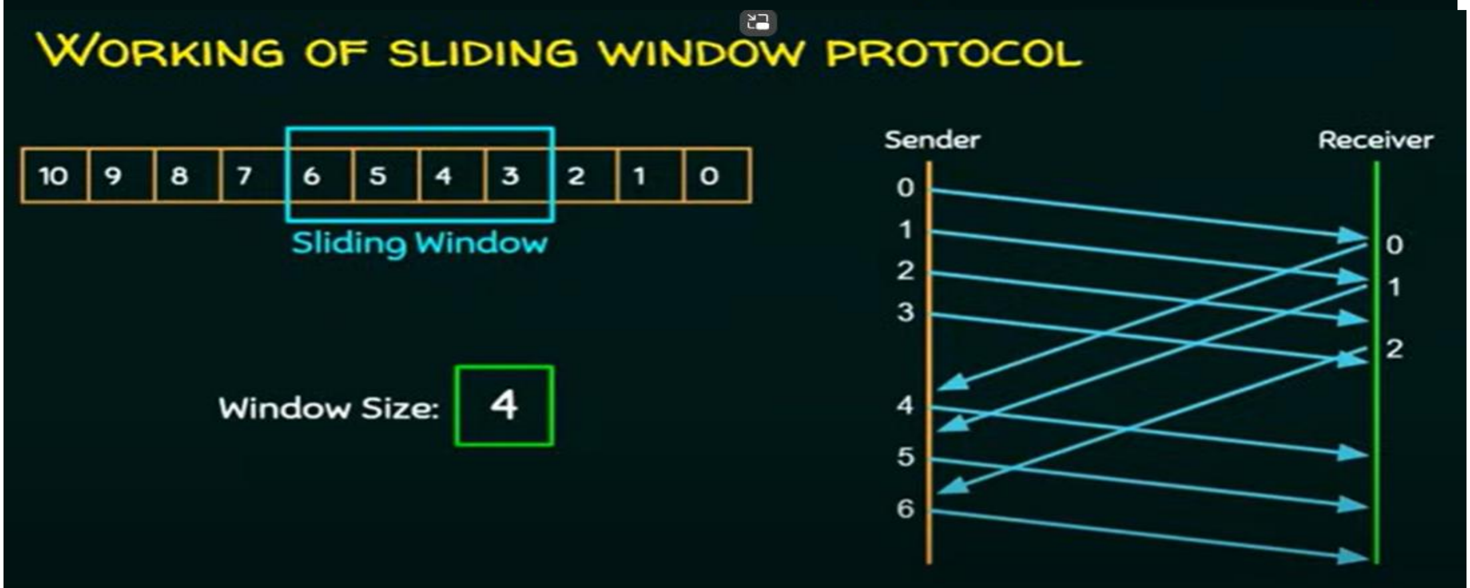
Working Principle:

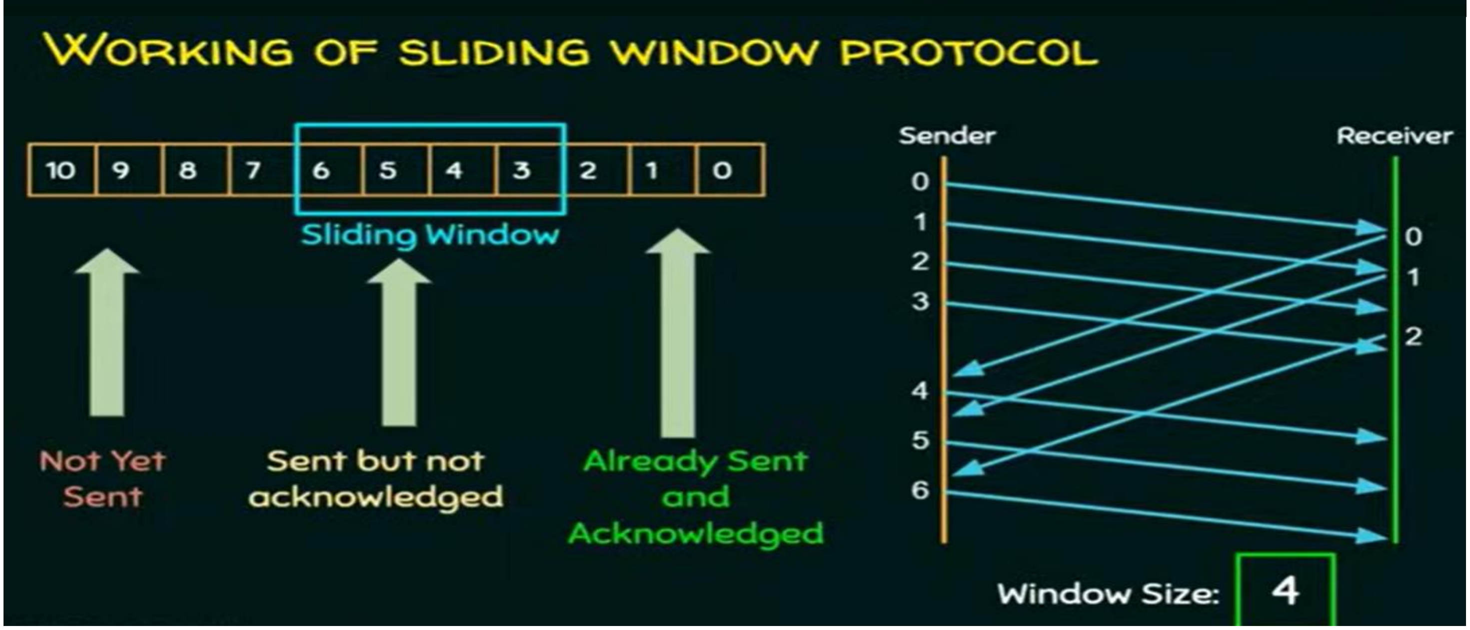
In these protocols, the sender has a buffer called the sending window and the receiver has buffer called the receiving window.

The sequence numbers are numbered as modulo-n. For example, if the sending window size is 4, then the sequence numbers will be 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, and so on. The number of bits in the sequence number is 2 to generate the binary sequence 00, 01, 10, 11.




WORKING OF SLIDING WINDOW PROTOCOL
10 9 8 7 6 5 4 3 2 1 0
Sliding Window
Window Size: 4
Sender
Receiver
0 1 2 3 4
0


WORKING OF SLIDING WINDOW PROTOCOL
10 9 8 7 6 5 4 3 2 1 0
Sliding Window
Window Size: 4
Sender
Receiver
0 1 2 3 4 5 6
0 1 2


WORKING OF SLIDING WINDOW PROTOCOL
10 9 8 7 6 5 4 3 2 1 0
Sliding Window
Not Yet Sent
Sent but not acknowledged
Already Sent and Acknowledged
Sender
Receiver
0 1 2 3 4 5 6
0 1 2
Window Size: 4

Stop Wait ARQ Sliding Window Protocols:

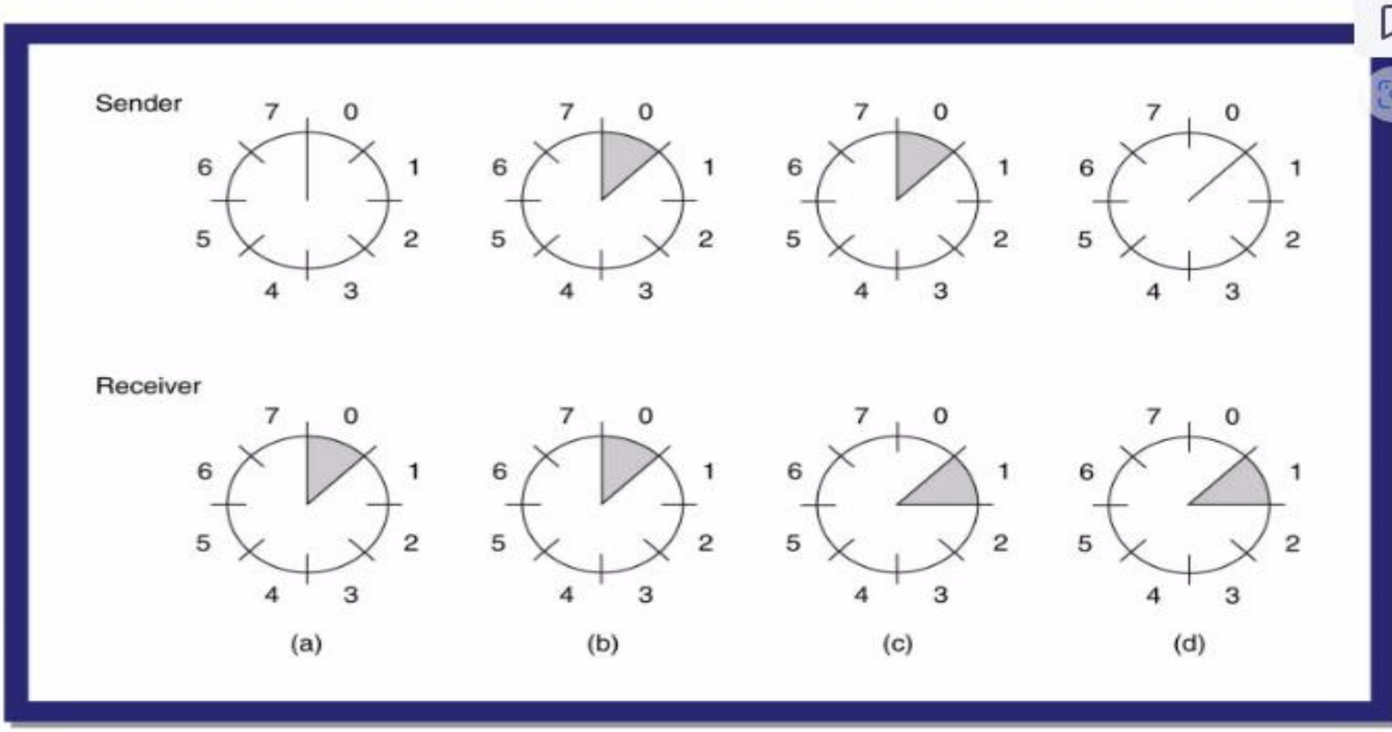
- 1-bit Sliding Window
- Sliding window using Go Back N
- Sliding Window Using Selective Repeat

One – bit Sliding window protocol(one bit):

In one – bit sliding window protocol, the size of the window is 1. So the sender transmits a frame, waits for its acknowledgment, then transmits the next frame. Thus it uses the concept of stop and waits for the protocol.

Working Principle

- The data frames to be transmitted additionally have an acknowledgment field, ack field that is of a few bits length.
- The ack field contains the sequence number of the last frame received without error.
- If this sequence number matches with the sequence number of the frame to be sent, then it is inferred that there is no error and the frame is transmitted.



-A sliding window of size 1, with a 3-bit sequence number-

(a) Initially.
(b) After the first frame has been sent.
(c) After the first frame has been received.
(d) After the first acknowledgement has been received.

→ The above example has a sliding window of size 1, with a 3-bit sequence number

↳ sequence number range is 0 to $2^3 - 1$

$= 0$ to 7

(a) Initially

(b) After the first frame has been sent

(c) After the first frame has been received

(d) After the first Ack has been received.

(a) Sender :- Initially when data transmission is not yet started.

Receiver :- waiting for a frame of sequence num = 0.

(b) Sender :- Sender sends a frame of sequence num = 0

Receiver :- waiting for a frame of sequence num = 0

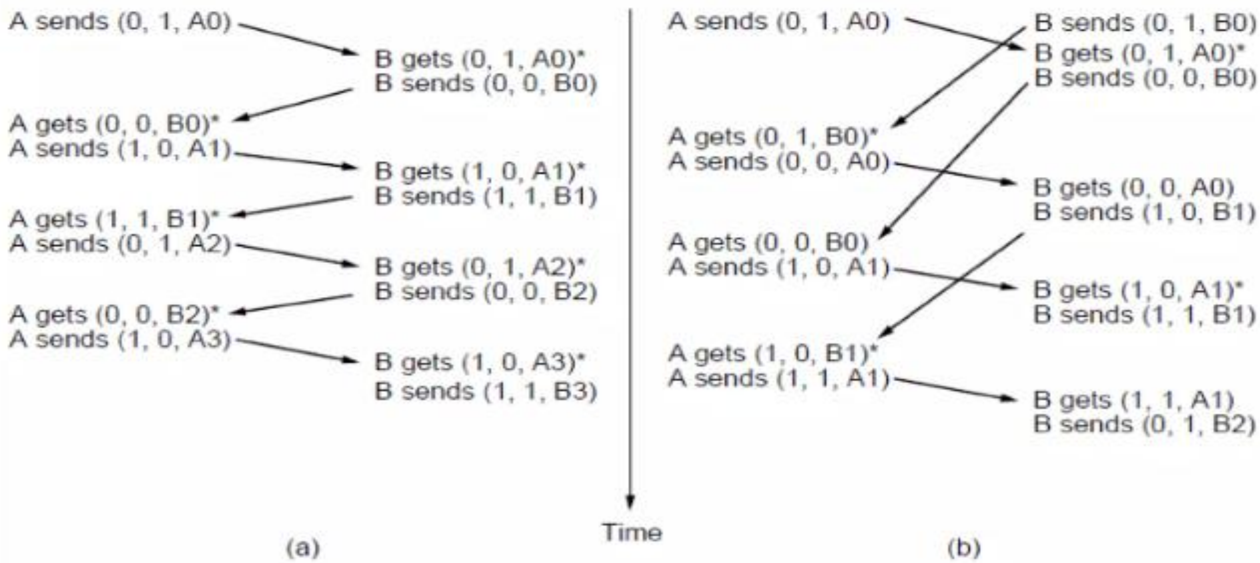
(c) Sender :- Sender waiting for an Ack for frame of seq num = 0

Receiver :- sends the Ack and waits for the next frame of seq num = 1

(d) Sender :- Data transmission of frame with seq num = 0 is completed & not sending any data (idle).

Receiver : waiting for a frame of seq num = 1.

One bit Sliding Window protocol



- The notation is (seq, ack, packet number).

Part (a): If B waits for A's first frame before sending one of its own. Each frame arrival brings a new packet for the network layer; there are no duplicates.

Part (b): If A and B simultaneously initiate communication, their first frames cross, and the data link layers gets into a situation. Half of the frames contain duplicates, even though there are no transmission errors

G0 BACK N ARQ:

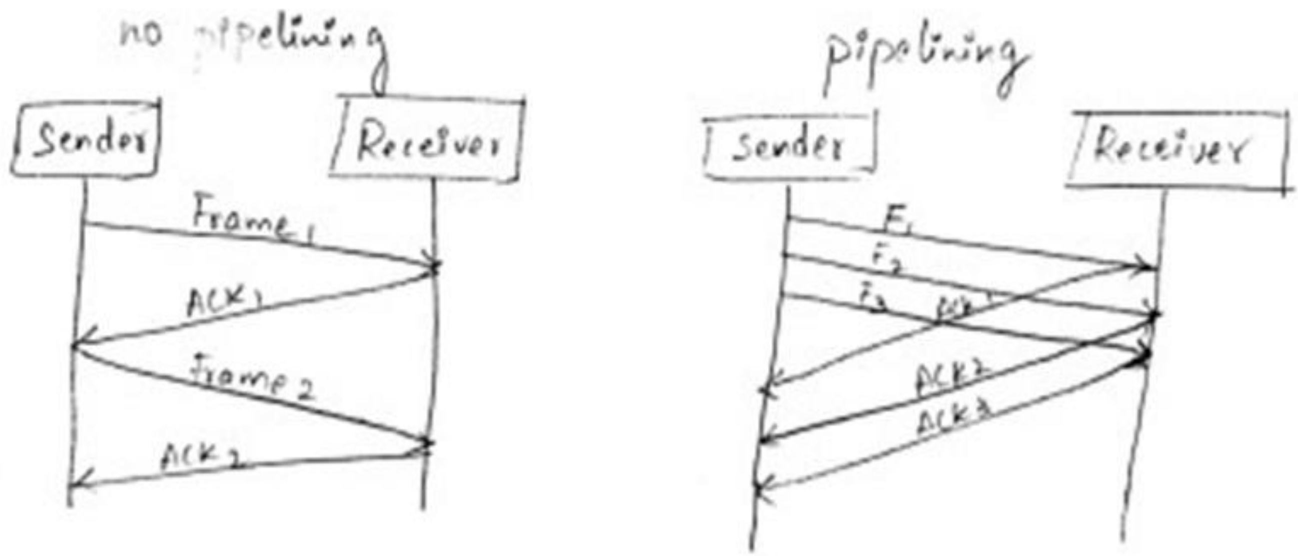
Go–Back–N ARQ

- Go – Back – N ARQ uses the concept of protocol pipelining i.e. the sender can send multiple frames before receiving the acknowledgment for the first frame.
- There are finite number of frames and the frames are numbered in a sequential manner.
- The number of frames that can be sent depends on the window size of the sender.
- If the acknowledgment of a frame is not received within an agreed upon time period, all frames in the current window are transmitted.

Sender window size of Go-Back-N Protocol is N.

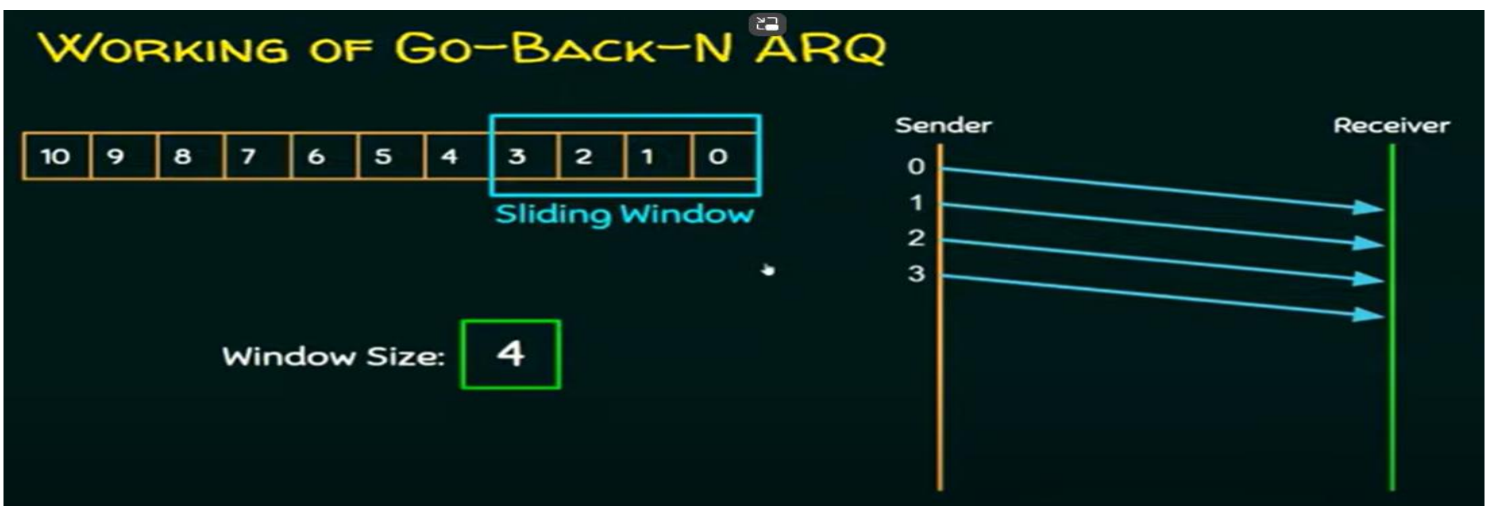
Receiver window size of Go-Back-N Protocol is 1.

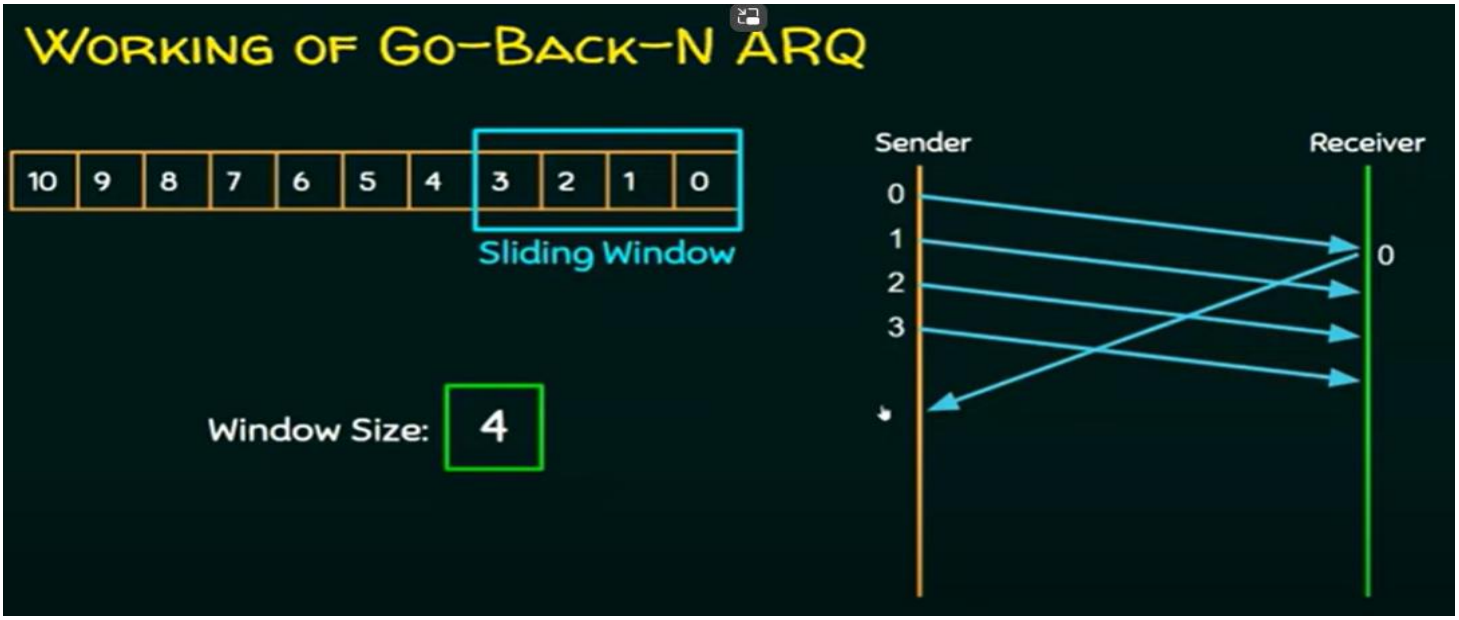
pipelining :- It is a technique in which multiple frames are sent at a time without waiting for the corresponding individual acknowledgements.

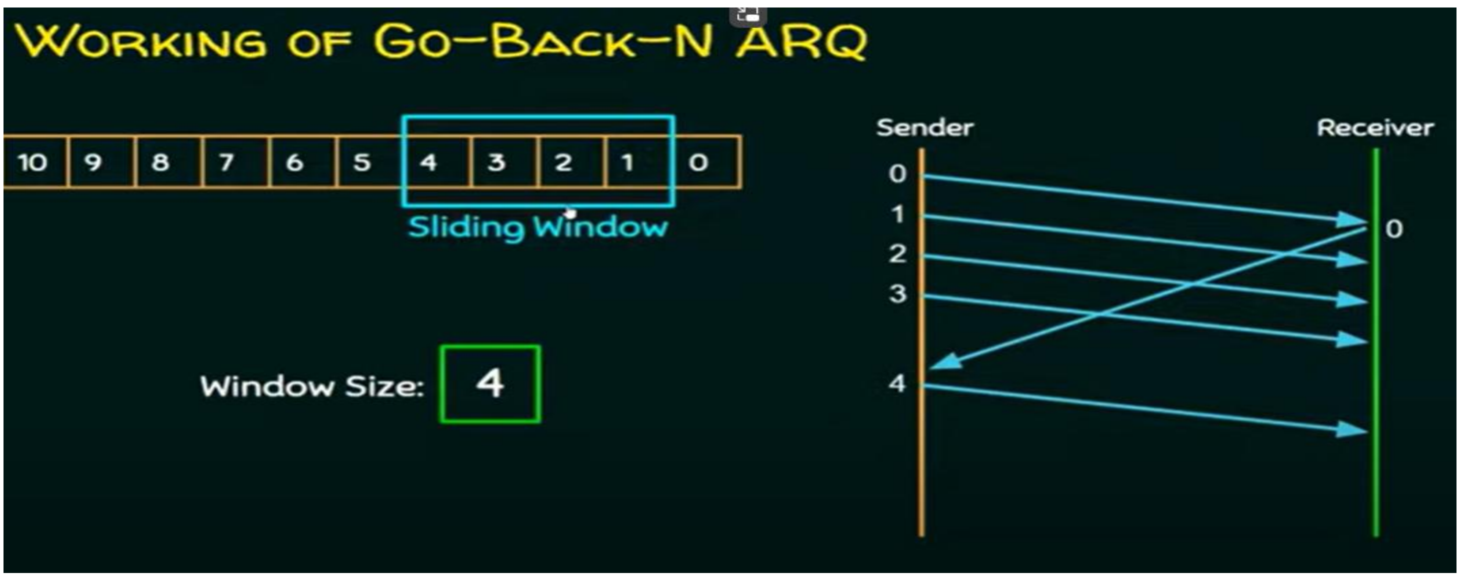


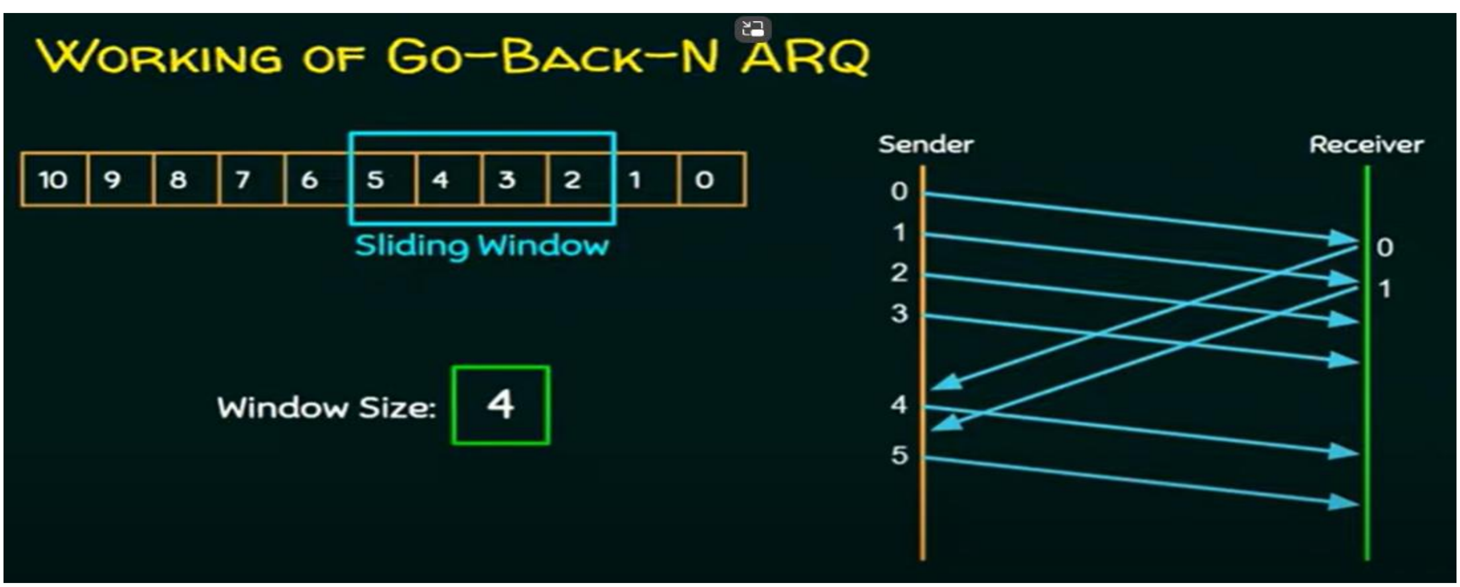
- In this protocol, the sender retransmits all the frames that are transmitted after the damaged/lost frame.
- It error rate is high, it wastes a lot of bandwidth.
- In this protocol, the receiver do not store the frames received after the damaged frame until the damaged frame is retransmitted.
- It is a mechanism to detect & control the errors.
- Go-Back-N protocol is shown in the below diagram.
- Frame 2 is lost, so all the frames followed by frame 2 are deleted (discarded).

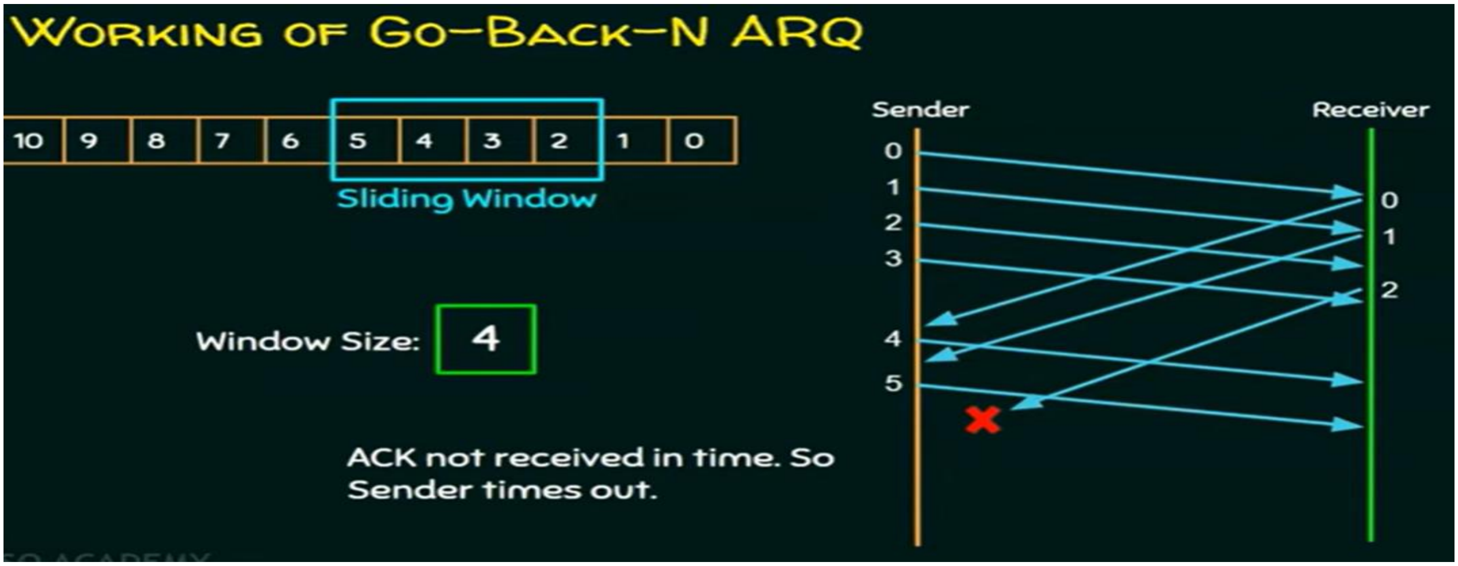
➢ All the frames from frame 2 to frame 5 are re transmitted




WORKING OF GO-BACK-N ARQ
10 9 8 7 6 5 4 3 2 1 0
Sliding Window
Window Size: 4
Sender
Receiver
0
1
2
3
0


WORKING OF GO-BACK-N ARQ
10 9 8 7 6 5 4 3 2 1 0
Sliding Window
Window Size: 4
Sender
Receiver
0
1
2
3
4
0


WORKING OF GO-BACK-N ARQ
10 9 8 7 6 5 4 3 2 1 0
Sliding Window
Window Size: 4
Sender
Receiver
0
1
2
3
4
5
0
1



When the receiver finds an error in frame, it tells the transmitter to go back, resend that frame and all succeeding frames. This feature guarantees that the frame is received in order, but it also means that some good frames may have to be resent.

Advantages of Go back N:

- The sender can send as many frames at a time
- One ACK can acknowledge one or more frame
- Efficiency is more
- Waiting time is low

We can alter the size of the sender window

Disadvantages of Go Back N ARQ:

- Buffer is required
- Sender needs to duplicate the last send N frames.
- Retransmission of many error free packet because of resending the nth error packet

SELECTIVE REPEAT ARQ:

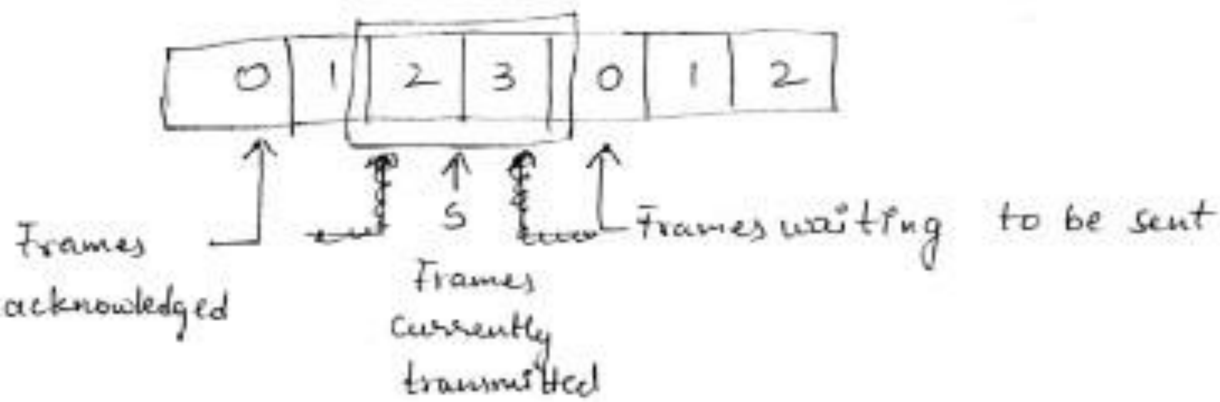
(c) A protocol using selective repeat.

- The go-back-n protocol works well if errors are rare, but if the channel has high error rate, it wastes lot of bandwidth on retransmitted frames.
- An alternative approach is selective repeat.
- The selective repeat protocol retransmits only that frame which is damaged or lost.
- The sender maintains a buffer (Sender window) having the

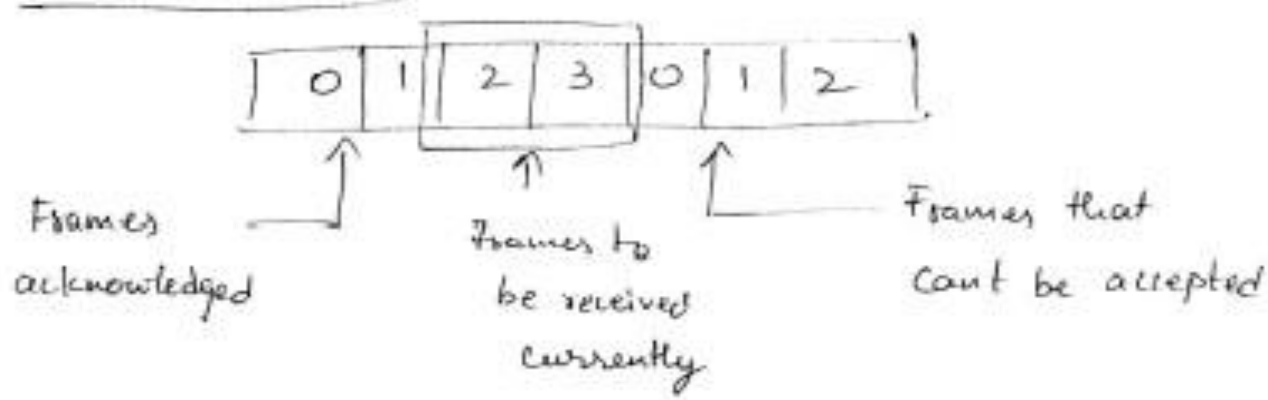
sequence numbers of the frames that the sender is allowed to send.

- The receiver also maintains a buffer (receiver window) having the sequence numbers of the frames that the receiver is allowed to receive.
- In this protocol, receiver stores the frames received after the damaged frame in the buffer until the damaged frame is replaced.

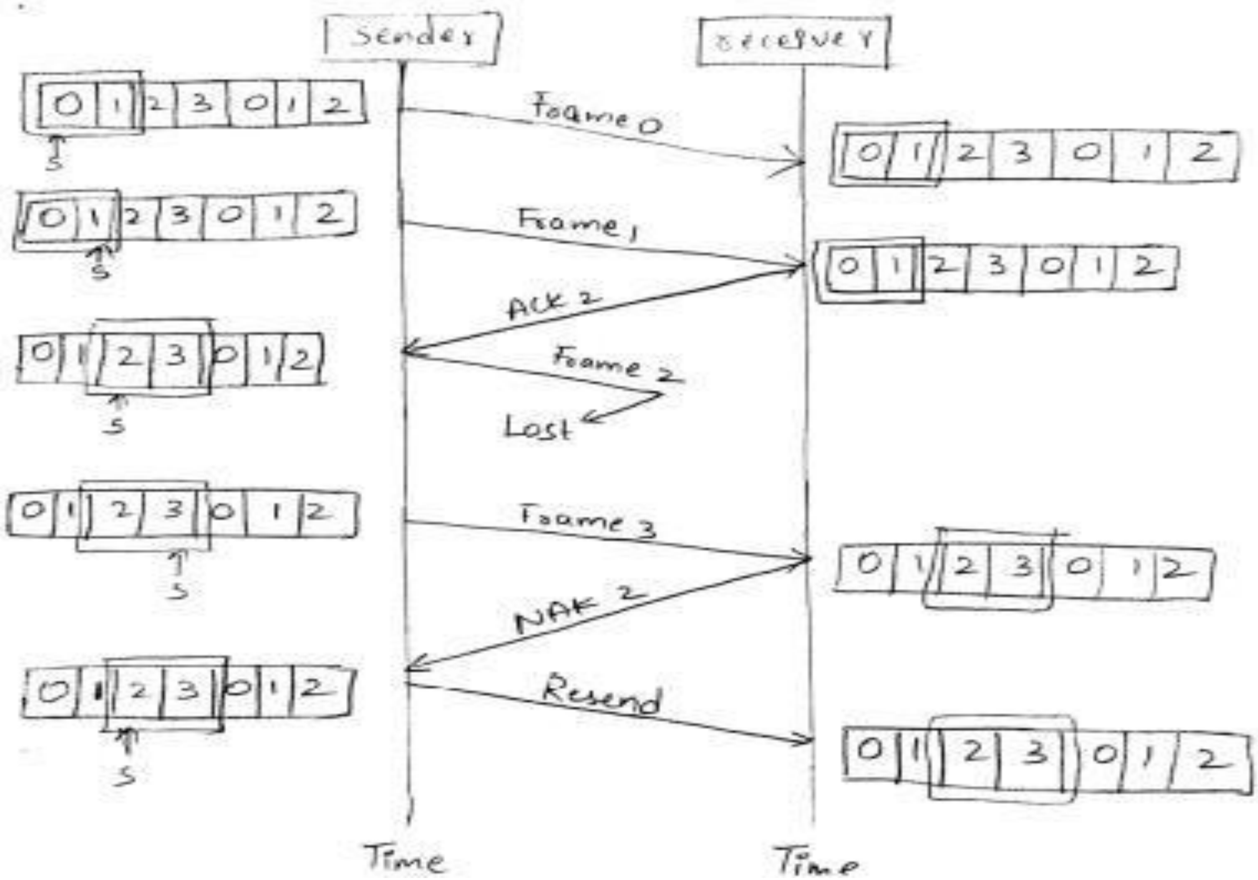
Sender window :



receiver window :



∴ The selective repeat protocol is shown in the below example. The Frame 2 is lost, only that frame is retransmitted.



Data Link Layer Protocol in HDLC

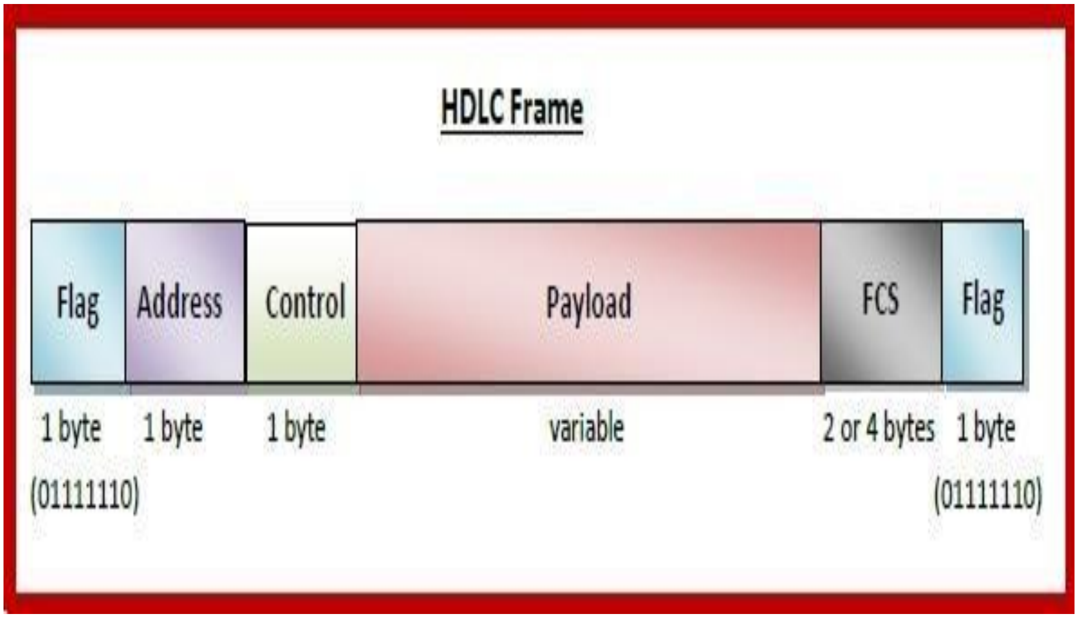
High-level Data Link Control (HDLC) is a group of communication protocols of the data link layer for transmitting data between network points or nodes. Since it is a data link protocol, data is organized into frames. A frame is transmitted via the network to the destination that verifies its successful arrival.

It is a bit - oriented protocol that is applicable for both point - to - point and multipoint communications.

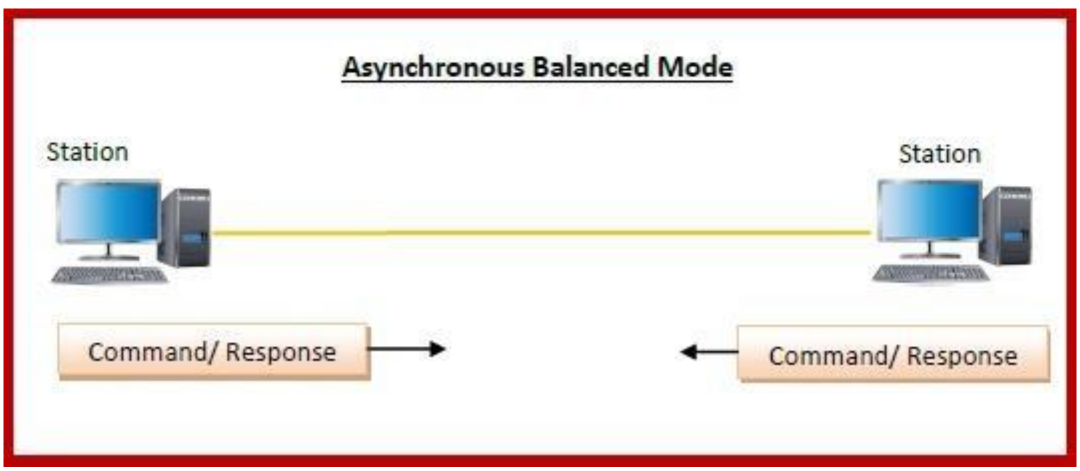
Transfer Modes

HDLC supports two types of transfer modes, normal response mode and asynchronous balanced mode.

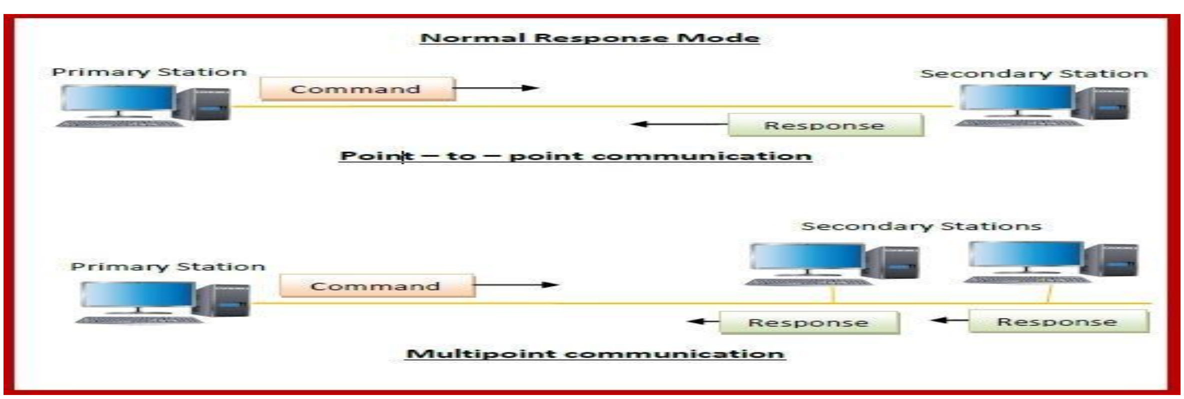
Normal Response Mode (NRM) – Here, two types of stations are there, a primary station that send commands and secondary station that can respond to received commands. It is used for both point - to - point and multipoint communications.



Asynchronous Balanced Mode (ABM) – Here, the configuration is balanced, i.e. each station can both send commands and respond to commands. It is used for only point - to - point communications.



HDLC Frame Structure



HDLC is a bit - oriented protocol where each frame contains up to six fields. The structure varies according to the type of frame. The fields of a HDLC frame are

Flag – It is an 8-bit sequence that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.

Address – It contains the address of the receiver. If the frame is sent by the primary station, it contains the address(es) of the secondary station(s). If it is sent by the secondary station, it contains the address of the primary station. The address field may be from 1 byte to several bytes.

Control – It is 1 or 2 bytes containing flow and error control information.

Payload – This carries the data from the network layer. Its length may vary from one network to another.

FCS – It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

Types of HDLC Frames

There are three types of HDLC frames. The type of frame is determined by the control field of the frame

I-frame – I-frames or **Information frames** carry user data from the network layer. They also include flow and error control information that is piggybacked on user data. The first bit of control field of I-frame is 0.

S-frame – S-frames or **Supervisory frames** do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bit of control field of S-frame is 10.

U-frame – U-frames or **Un-numbered frames** are used for myriad miscellaneous functions, like link management. It may contain an information field, if required. The first two bit of control field of U-frame is 11.

Point to point protocol (PPP)

- The PPP stands for Point-to-Point protocol. It is the most commonly used protocol for point-to-point access. Suppose the user wants to access the internet from the home, the PPP protocol will be used.
- The PPP protocol can be used on synchronous link like ISDN as well as asynchronous link like dial-up. It is mainly used for the communication between the two devices.
- It is also known as a byte-oriented protocol

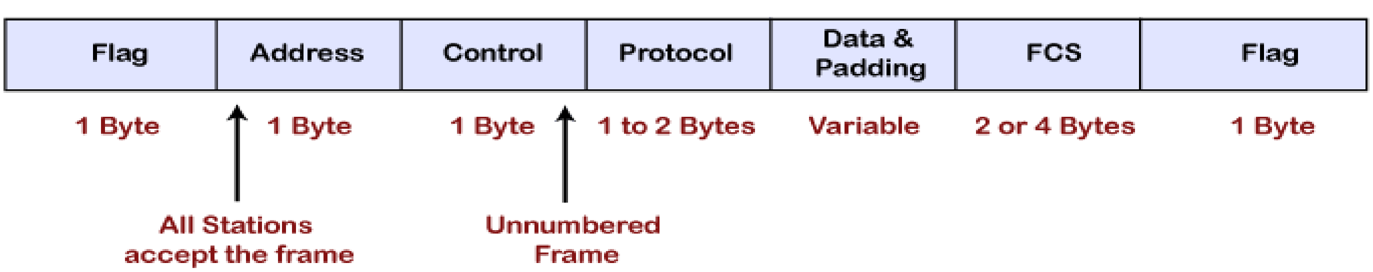
Services provided by PPP

- It defines the format of frames through which the transmission occurs.
- It defines the link establishment process.
- It defines data exchange process, i.e., how data will be exchanged, the rate of the exchange.
- The main feature of the PPP protocol is the encapsulation. It defines how network layer data and information in the payload are encapsulated in the data link frame.
- It defines the authentication process between the two devices. The authentication between the two devices, handshaking and how the password will be exchanged between two devices are decided by the PPP protocol.

Services not provided by the PPP protocol(RFC1661)

- It does not support flow control mechanism.
- It has a very simple error control mechanism.
- As PPP provides point-to-point communication, so it lacks addressing mechanism to handle frames in multipoint configuration.

Frame format of PPP protocol



Flag: The flag field is used to indicate the start and end of the frame. The flag field is a 1-byte field that appears at the beginning and the ending of the frame. The pattern of the flag is similar to the bit pattern in HDLC, i.e., 01111110.

Address: It is a 1-byte field that contains the constant value which is 11111111. These 8 ones represent a broadcast message.

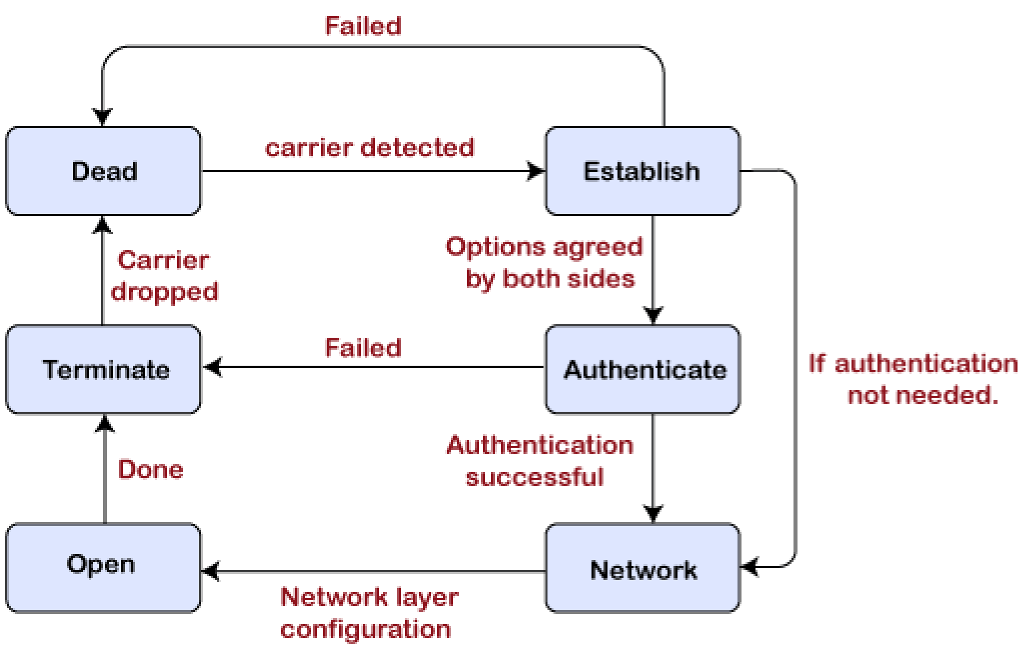
Control: It is a 1-byte field which is set through the constant value, i.e., 11000000. It is not a required field as PPP does not support the flow control and a very limited error control mechanism. The control field is a mandatory field where protocol supports flow and error control mechanism.

Protocol: It is a 1 or 2 bytes field that defines what is to be carried in the data field. The data can be a user data or other information.

Payload: The payload field carries either user data or other information. The maximum length of the payload field is 1500 bytes

CRC: It is a 16-bit field which is generally used for error detection

Transition phases of PPP protocol



Transition phases

Dead: Dead is a transition phase which means that the link is not used or there is no active carrier at the physical layer.

Establish: If one of the nodes starts working then the phase goes to the establish phase. In short, we can say that when the node starts communication or carrier is detected then it moves from the dead to the establish phase.

Authenticate: It is an optional phase which means that the communication can also moves to the authenticate phase. The phase moves from the establish to the authenticate phase only when both the communicating nodes agree to make the communication authenticated.

Network: Once the authentication is successful, the network is established or phase is network. In this phase, the negotiations of network layer protocols take place.

Open: After the establishment of the network phase, it moves to the open phase. Here open phase means that the exchange of data takes place. Or we can say that it reaches to the open phase after the configuration of the network layer.

Terminate: When all the work is done then the connection gets terminated, and it moves to the terminate phase.

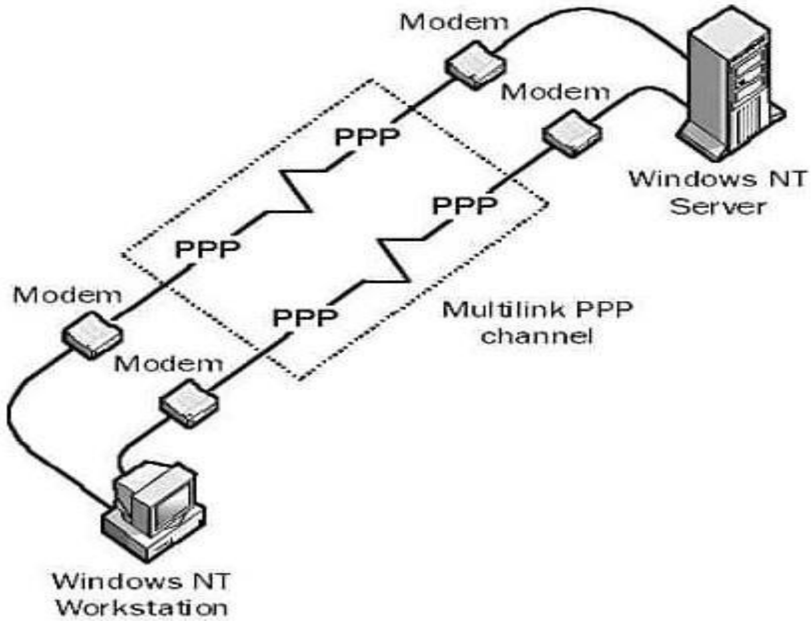
Components in PPP

- Link Control Protocol (LCP)
- Authentication protocols

Link Control Protocol (LCP): The role of LCP is to establish, maintain, configure, and terminate the links. It also provides negotiation mechanism.

Authentication protocols: This protocol plays a very important role in the PPP protocol because the PPP is designed for use over the dial-up links where the verification of user identity is necessary. Thus this protocol is mainly used to authenticate the endpoints for the use of other services.

Multilink PPP



Multilink PPP (also referred to as **MLPPP**, **MP**, **MPPP**, **MLP**, or Multilink) provides a method for spreading traffic across multiple distinct PPP connections. It is defined in RFC 1990. It can be used, for example, to connect a home computer to an Internet Service Provider using two traditional 56k modems, or to connect a company through two leased lines.

On a single PPP line frames cannot arrive out of order, but this is possible when the frames are divided among multiple PPP connections. Therefore, Multilink PPP must number the fragments so they can be put in the right order again when they arrive.

Multilink PPP is an example of a link aggregation technology. Cisco IOS Release 11.1 and later supports Multilink PPP.

Multilink PPP is a communications strategy that makes use of the basic concept of **point-to-point protocol**. Essentially, the approach allows for the utilization of more than one PPP communication port in order to achieve a higher amount of **bandwidth** to work with. This type of **communications protocol** can often be employed with a personal computer and thus enhance the overall efficiency of many tasks The utilization of Multilink PPP can be especially helpful in locations where dial up connections to the Internet are the only alternative. The end user will make use of two different modems to establish independent connections to the Internet. The connections are made to the same Internet **Service Provider**. Assuming that the **ISP** allows for this type of connectivity, the end user can effectively double the operating speed, as the two connections divide the requested data into packets that are simultaneously transmitted through each connection, then recombined at the user end.

However, there are other applications for Multilink PPP that go beyond simply increasing the speed associated with dial up service. The same concept is often employed in fiber optic systems that function primarily as a means of providing audio signaling. Cable modems can also employ Multilink PPP to enhance transmission. Satellite transmissions can also make use of the principles of Multilink PPP in order to increase the efficiency of data transfers. The overall simplicity of employing this type of multiple connection protocol makes it easily adaptable to a number of different situations, and can often be an ideal solution for persons located in relatively isolated areas.